

Practical 11 : 1.Create databases and tables, insert small amounts of data, and run simple queries using Impala

-- Create and use database

```
CREATE DATABASE IF NOT EXISTS student_db;  
USE student_db;
```

-- Create tables

```
CREATE TABLE students (  
    student_id INT,  
    name STRING,  
    age INT,  
    department STRING  
);
```

```
CREATE TABLE courses (  
    course_id INT,  
    course_name STRING,  
    instructor STRING  
);
```

```
CREATE TABLE enrollments (  
    enroll_id INT,  
    student_id INT,  
    course_id INT,  
    marks INT  
);
```

-- Insert data into students

```
INSERT INTO students VALUES  
(1, 'Amit', 20, 'Computer'),  
(2, 'Neha', 21, 'IT'),  
(3, 'Rahul', 22, 'Electronics'),  
(4, 'Sneha', 20, 'Computer');
```

-- Insert data into courses

```
INSERT INTO courses VALUES
```

```
(101, 'DBMS', 'Dr. Sharma'),
```

```
(102, 'Data Structures', 'Prof. Mehta'),
```

```
(103, 'Networks', 'Dr. Khan');
```

```
-- Insert data into enrollments
```

```
INSERT INTO enrollments VALUES
```

```
(1, 1, 101, 85),
```

```
(2, 1, 102, 78),
```

```
(3, 2, 101, 88),
```

```
(4, 3, 103, 90),
```

```
(5, 4, 102, 75);
```

```
-- View all students
```

```
SELECT * FROM students;
```

```
-- Filter students (age > 20)
```

```
SELECT * FROM students
```

```
WHERE age > 20;
```

```
-- Students from Computer department
```

```
SELECT name FROM students
```

```
WHERE department = 'Computer';
```

```
-- Join students and enrollments
```

```
SELECT s.name, e.marks
```

```
FROM students s
```

```
JOIN enrollments e
```

```
ON s.student_id = e.student_id;
```

```
-- Join all tables
```

```
SELECT s.name, c.course_name, e.marks
```

```
FROM students s
```

```
JOIN enrollments e ON s.student_id = e.student_id
```

```
JOIN courses c ON e.course_id = c.course_id;
```

-- Average marks

```
SELECT AVG(marks) AS average_marks  
FROM enrollments;
```

-- Highest marks

```
SELECT MAX(marks) AS highest_marks  
FROM enrollments;
```